ABC Vehicraft: The Aerial Boat Car, and What a Consumer EEG Headband Records While Riding It

Alexander Vicol and Steve Mann
University of Toronto and MannLab Canada, Toronto, Canada

***Abstract*—We introduce the ABC Vehicraft, a craft that is at once aerial, boat and car, and report the first brain-sensing recordings taken while riding it. We tested three designs, each carrying its own helium balloon cluster as the aerial element: a paddle board with balloons, a water-walking ball with balloons, and the Billy boat with balloons, a small inflatable ridden in the water with the ears at the waterline. On one afternoon in Toronto's harbour the second author rode all three wearing a Muse S Athena headband streaming to a phone. No headset, no laptop, no marker, 3.6 to 4.9 minutes per mode. We ask what such a recording is good for. The paddle board gives usable EEG on 88% of 2 s epochs behind the ears and 98% on the left forehead; the ball halves that and the Billy boat leaves one epoch in six behind the ears. The theta/beta ratio, the attentional-control marker of our State-of-Float work, reads 2.7 on the paddle board, 1.6 in the Billy boat, and 6.9 in the ball, where a 2.4 Hz step rhythm in the accelerometer and a threefold rise in delta and theta say that the number is motion, not mind. The headband's own arousal index sat at 0.41, 0.49 and 0.47. The optical channels logged nothing, so there is no fNIRS. The paper's contribution is the craft and an honest account of what a wet, moving, open-air headband gives, with the recording rules that follow for the next ride: a strap over the band, dry ear sites, a logged fNIRS stream, a trigger per mode, and two-minute eyes-open and eyes-closed blocks in each.**

***Index Terms*—Mersivity; wearable computing; EEG; brain-computer interface; consumer electronics; water-walking ball; paddle board; theta/beta ratio; amphibious vehicle; extended reality.**

I. Introduction

Mersivity is the idea that technology should connect people to each other and to the natural world rather than wall them off from it [1], and that wearable sensing [2] lets the laboratory come out to the lake instead of bringing the lake's people into a laboratory. Our State-of-Float paper took a consumer EEG headband into a water-walking ball and found a higher theta/beta ratio while floating than on land [1]. That work was done lying still under a VR headset, in two-minute blocks, with the headset pressing the headband into place.

This paper goes the other way. The craft moves, the rider stands, walks or swims, nothing presses the headband down, and the water reaches the electrodes. The question is what a consumer headband records under those conditions, because the answer decides whether brain-state measurement can travel with a vehicle at all. We do it on the ABC Vehicraft, a craft built to be aerial, boat and car at once, and we record its rider on three of its modes in one afternoon.

We make no claim about the rider's state from these recordings. Three to five minutes per mode, one person, no markers and no baseline is not a study of mind. It is a study of the measurement, and it yields a table of what each mode gives, which artifacts dominate, and what the next ride must add. That is the contribution, together with the craft itself.

II. Background

Mersivity and the water. This paper belongs to a programme that takes sensing into water and play rather than bringing people into a laboratory [1]. Its instruments include the hydraulophone, whose keys are water jets [3], and the wider family of physiphones named for the matter that makes their sound [4]; its brain sensing runs on consumer headbands, from the eye as a camera [5] and sleep staging with the adaptive chirplet transform [6] to the cold-plunge game Cocaine Buzzkill [7] and State-of-Float, which found a higher theta/beta ratio floating in a water ball than on land [1]. Our recording app MuseLog [8] is the common tool. Wearable computing as Mann framed it keeps the person in the loop and in control [2], [9], and the record belongs to the person who makes it [10].

Sensing in the field. Fletcher's work set the standard for taking physiology out of the clinic. iCalm logged autonomic arousal from a wrist sensor through a whole day [11]; the same group built a wearable and phone application for therapy in addiction and PTSD [12], studied craving in recovery with iHeal [13], measured screen exposure from a wearable [14] and took mobile tools to people experiencing homelessness [15]. Two lessons run through it: low-cost sensors reach places health care does not, but only when designed for the conditions



Fig. 1. The ABC Vehicraft in Toronto's inner harbour, October 9, 2026: the paddle-board design with its balloon cluster, the city behind. The rider wears a consumer EEG headband.



Fig. 2. Design 1, the paddle board with balloons, in Toronto's inner harbour on October 9, 2026: under way with the cluster at the bow. The rider wears the Muse S Athena headband.



Fig. 3. Design 2, the ball with balloons: the rider walks inside the water-walking ball, the balloon cluster tethered alongside on a line.



Fig. 4. Design 3, the Billy boat with balloons: the small yellow inflatable held at the chest, ridden swimming with the ears at the waterline under the cluster, at sunset.

they meet [16], [17], and a measurement's failure rate across people and settings is itself a result [18]. Affective wearables were proposed for exactly this kind of everyday, in-context sensing [19]. A vehicle on open water is an extreme of "the conditions they meet", and that is why we measure the measurement.

III. The ABC Vehicraft

The ABC Vehicraft is one idea built three ways: a craft that belongs to the air, the water and the land at once, made from parts anyone can carry to a shore. Each design joins the three elements. *Aerial*: a cluster of helium foil and latex balloons tethered to the craft. It lifts only a small fraction of the load, so it is the aerial element in form and the start of it in function: it makes the craft visible from shore and from the water, a safety flag that cannot be lowered by accident, and it is the part that a larger envelope would turn into real lift. *Boat*: whatever floats the rider, a board, a ball or an inflatable. *Car*: whatever rolls, a ball that rolls on land and on water alike, or a board set on wheels. The craft is for play in the open air and water, which is where Mersivity places technology [1], and for carrying wearable sensing out of the laboratory with the rider.

We tested three designs (Figs. 2 to 4).

1. Paddle board with balloons. A stand-up paddle board, inflatable, three metres long, ridden standing with a paddle; the balloon cluster, gold foil letters and black and gold latex spheres, is tied at the bow. It is the most boat-like of the three and the steadiest platform for the rider.

2. Ball with balloons. A water-walking ball, a transparent inflatable sphere two metres across that the rider walks inside [1], [20], with the balloon cluster tethered alongside on a line. It is the design that behaves like a wheel: the same ball rolls on land and on water, so it is car and boat in one hull, and its motion is the rider's walking.

3. Billy boat with balloons. A small yellow inflatable, the Billy boat, that the rider holds at the chest and rides through the water swimming, the head low and the ears at the waterline, with the balloon cluster standing over it. It is the smallest and most amphibious design, the rider half in the water, and the hardest on a headband.

The car element also stands on its own: a board set on skateboard wheels carries a rider on land with the same balloons in hand (Fig. 11), so the craft can roll from the street to the water's edge. Figs. 1, 2 to 4 and 12 show the three water designs on October 9, 2026, in Toronto's inner harbour; the rider wears a Muse S Athena headband throughout.

Each design was recorded as one continuous MuseLog file (Table I): standing and paddling on the paddle board at 17:51, walking inside the ball at 18:16, and riding the Billy boat at 18:51, with the ears at or under the surface.

IV. Methods

Recording. Muse S Athena, four dry EEG electrodes at TP9, AF7, AF8 and TP10 referenced at Fpz, 256 Hz, streamed over Bluetooth to a phone running MuseLog, which writes one CSV row per sample with the raw channels, the headband's own band powers, head-set indicators, accelerometer and gyroscope, the optical channels, and the headband's arousal index. No headset, no laptop, no stimulus. The headband's fit indicator read 4, its poorest value, on every channel for the whole of every file; what that meant in practice is what the usable-epoch counts below measure. The optical columns (730

TABLE I

THE THREE MODES AND WHAT EACH RECORDING GAVE. USABLE EPOCHS ARE 2 S EPOCHS UNDER 200 μV ON THE EAR CHANNELS (TP9, TP10) AND ON AF7. TBR IS THE MEDIAN THETA/BETA RATIO OVER 10 S WINDOWS. RHYTHM IS THE ACCELEROMETER'S DOMINANT FREQUENCY.

mode	start	min	ears / AF7 usable	TBR	arousal	rhythm
paddle board, standing	17:51	4.9	88% / 98%	2.7	0.41	2.3 Hz
water ball, walking	18:16	3.6	50% / 72%	6.9	0.49	2.4 Hz
Billy boat, ears in water	18:51	3.6	16 to 33% / 93%	1.6	0.47	1.05 Hz



Fig. 5. The first author on the same craft earlier in the afternoon, the balloon cluster at the bow, in the inner harbour.

and 850 nm, red and ambient, inner and outer, both sides) are empty in all three files: the fNIRS stream was not logged, so there is no fNIRS in this paper.

EEG. Raw channels band-passed 1 to 100 Hz, cut into 2 s epochs, an epoch usable when its peak stays under 200 μV. Spectra are Welch estimates over the usable epochs (4 s segments). Band powers are delta 1 to 4, theta 4 to 8, alpha 8 to 13, beta 13 to 30 and gamma 30 to 45 Hz. The theta/beta ratio (TBR), the marker of our State-of-Float work and a marker of attentional control [21], [22], is theta over beta per 10 s window, the median over the four channels, usable windows only. We report the headband's arousal index per 10 s as the device gives it.

Motion. The accelerometer magnitude's spectrum from 0.2 to 5 Hz gives the dominant rhythm of the head; the gyroscope magnitude's median gives the rotation rate.

V. RESULTS

Contact and artifact. Fig. 6. Standing on the paddle board the headband gave usable EEG on 88% of epochs behind both ears and 98% on the left forehead, at 30 μV rms behind the ears and 16 on AF7, numbers a quiet indoor session would not be ashamed of. The right forehead electrode (AF8) was off the skin in every mode, 11 to 22% usable, and is dropped from what follows. Walking in the ball halved the usable share behind the ears to 50% and on AF7 to 72%, with rms up by

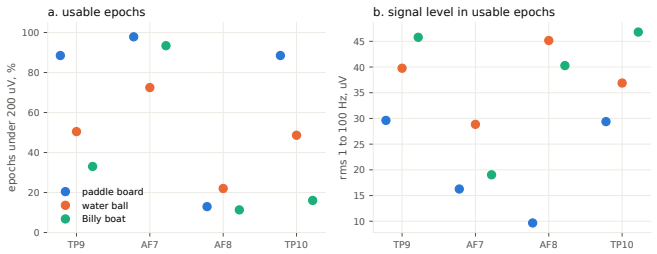


Fig. 6. What each mode gives per electrode: the share of 2 s epochs under 200 μV (left) and the rms of the usable epochs (right).

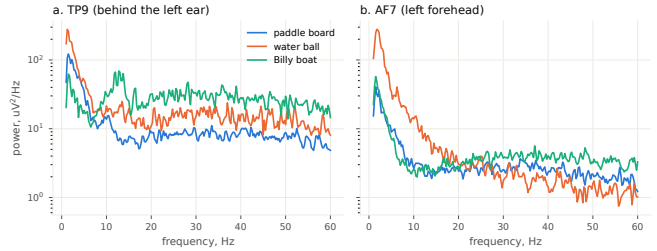


Fig. 7. Power spectra of the usable epochs, behind the left ear (left) and on the left forehead (right), one line per mode.

a third. With the ears in the water the ear electrodes gave one usable epoch in three on the left and one in six on the right, at 46 μV, while AF7, above the waterline, stayed 93% usable.

Spectra. Fig. 7. Behind the left ear the paddle board's spectrum is the ordinary 1/f shape with an alpha shoulder near 10 Hz. The ball raises the whole spectrum by a factor of two to three and the low end by more. The Billy boat raises everything above 10 Hz by a factor of three to four with a broad hump at 12 to 16 Hz, the signature of water and muscle at the ear site rather than cortex. On the forehead the three modes differ mainly below 10 Hz, where the ball's walking rhythm and its harmonics sit.

Band powers and the theta/beta ratio. Fig. 8. On the ear channels the paddle board sits lowest in every band. The ball more than doubles delta and theta while adding less to beta, which is why its TBR is 6.9 against the paddle board's 2.7; the Billy boat adds most to alpha, beta and gamma, which is why its TBR is lowest at 1.6. Read as brain state these would be a calmer rider in the ball and an aroused one in the Billy boat. Read with the motion and contact figures they are the opposite of a finding: delta and theta inflated by a 2.4 Hz step rhythm and its harmonics in the ball, and beta and gamma inflated by water and muscle at the ear sites in the Billy boat. The TBR of the paddle board, the one mode where the electrodes held and the head moved little, is the only one of the three we would quote, and it is in the range our floating subjects gave on land [1].

Motion. Fig. 9. The accelerometer shows the rider's rhythm on each craft: a sharp 2.4 Hz line in the ball, which is the step rate of walking on a surface that rolls away underfoot, at a head-acceleration spread of 0.16 g and a median rotation

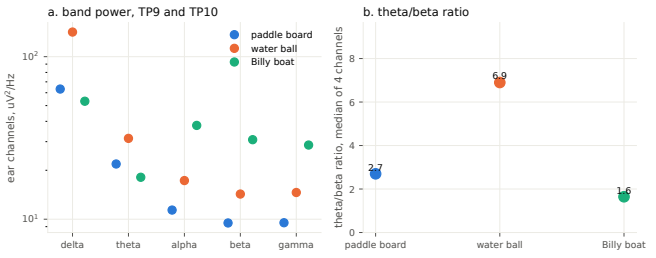


Fig. 8. Band power on the ear channels (left) and the theta/beta ratio (right) per mode. The ball's ratio is carried by motion-inflated delta and theta, the Billy boat's low ratio by water and muscle in the beta band.

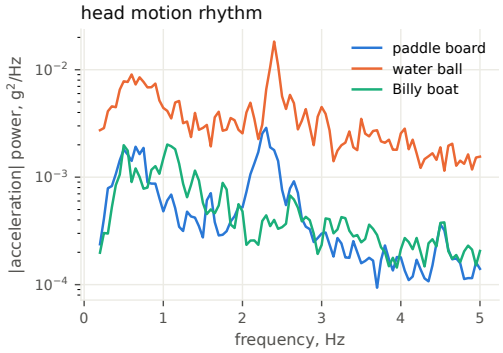


Fig. 9. Head-motion spectra from the headband's accelerometer. The ball's 2.4 Hz line is the rider's step rate.

rate of 22 degrees per second; a 2.3 Hz rhythm on the paddle board at 0.06 g, the board's chop and the paddle's cadence; and a 1.05 Hz rhythm in the Billy boat at 0.07 g, the swimming stroke. The step line in the ball sits inside the delta band's second harmonic and the theta band's third, which is where its EEG excess appears.

Through each ride. Fig. 10. On the paddle board the TBR falls through the first two minutes from 9 to about 2 and stays there, the pattern of a headband settling after it is put on wet rather than of a mind changing; the headband's arousal index is live for the first 1.7 minutes and then holds a constant 0.3, which is the device's output when it stops updating, not a reading. In the ball the TBR is measurable only in the usable windows and ranges from 2 to 17, and the arousal index swings between 0.1 and 0.85 with the rider's bouts of walking and standing. In the Billy boat the TBR settles from 5 to between 1 and 3 and the arousal index, available only from the second minute, sits at 0.3 to 0.7.

VI. What It Means

Which modes can be measured. Standing on a paddle board is a measurable condition for a consumer headband with nothing holding it down: nine epochs in ten usable behind the ears and the forehead nearly clean. Walking in the ball is a motion condition first and a brain condition second; its band powers are not interpretable without a motion model, and the honest number from it is the step rate. Ears in the water is not a

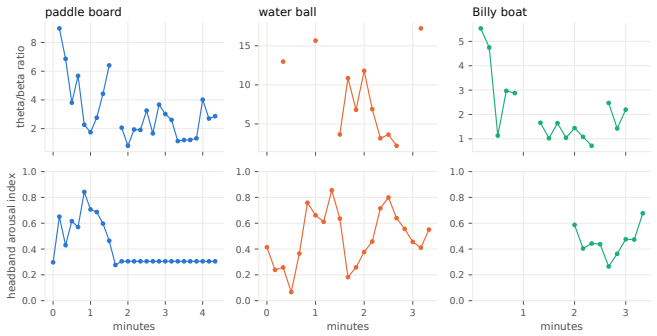


Fig. 10. The theta/beta ratio (top) and the headband's arousal index (bottom) per 10 s through each ride. Gaps are windows without a usable channel.



Fig. 11. The car element: a board on skateboard wheels with the balloons in hand, ridden by the first author on the harbour pier on October 17, 2026. This ride was not recorded with the headband.

condition the ear electrodes survive, though the forehead does, so a frontal-only analysis is possible there and an ear-based one is not.

The craft as a field instrument. Table II reads the three designs as three field conditions for a consumer sensor. The paddle board is the easy one and gave laboratory-quality EEG with nothing holding the headband; the ball is a motion condition, whose EEG is interpretable only with the accelerometer beside it; the Billy boat is a contact condition, where the forehead survives and the ears do not. Reporting those pass rates per design, rather than only the brain measures that survived them, is the practice the field-sensing literature asks for [18], and it tells the next rider which design to choose for which question. The whole kit, a headband (about US$250), a phone running MuseLog and a board, ball or inflatable, is carried to the shore by one person and run without a laptop, which is what makes the measurement repeatable by people who are not engineers [16].

Safety. Every ride was in a sheltered inner harbour beside

TABLE II

THE THREE DESIGNS SIDE BY SIDE: WHICH ELEMENTS EACH JOINS, HOW THE RIDER USES IT, AND WHAT THE HEADBAND GAVE ON IT. USABLE EPOCHS ARE 2 S EPOCHS UNDER 200 μV BEHIND THE EARS AND ON THE LEFT FOREHEAD (AF7).

design	boat	car	rider	head motion	usable, ears / AF7	what limits the EEG
paddle board with balloons	inflatable board	–	standing, paddling	2.3 Hz chop and stroke, 0.06 g	88% / 98%	nothing pressing the band down
ball with balloons	the ball	the ball rolls	walking inside	2.4 Hz steps, 0.16 g	50% / 72%	step rhythm in delta and theta
Billy boat with balloons	small inflatable	–	swimming, ears at waterline	1.05 Hz stroke, 0.07 g	16–33% / 93%	water at the ear electrodes
board on wheels (Fig. 11)	–	skateboard wheels	standing, rolling	not recorded	–	–



Fig. 12. The water-walking ball against the setting sun, with the balloon cluster tethered alongside and the tow line trailing to the camera.

a pier, with a personal flotation device on the board and in the Billy boat, a tow line on the ball and a second person on the pier. The water-walking ball is an air-tight enclosure and is used only for a few minutes at a time with the zip opened between rides [20]. The balloon clusters are tethered and recovered after every ride; none was released. Cold-water immersion raises catecholamines severalfold [23], so the Billy boat, the one design with the rider in the water, is the one to time and to watch.

Rules for the next ride. A strap or cap over the headband, which in our headset work improved contact by pressing the band in [1]. Dry the ear sites and keep them above the waterline when the ear channels matter. Log the optical stream: the Athena's fNIRS is the one signal that does not care about a wet electrode, and it was not recorded. A trigger press at the start of each mode and each change of activity, so the file carries its own timeline. Two-minute eyes-open and eyes-closed blocks in each mode, still, as in the State-of-Float protocol, so that the TBR has the baseline it needs. And a reference recording on land the same day. With those, the same afternoon becomes the study this paper is not.

Limits. One rider, one afternoon, three to five minutes per mode, no markers, no baseline, no fNIRS. The fit indicator read its poorest value throughout, so the usable-epoch rule is the only guard on contact. The arousal index is the device's own and its algorithm is not published. Nothing here says what riding the craft does to a mind; it says what the headband can see while a mind rides it.

VII. CONCLUSION

The ABC Vehicraft is a craft that is aerial, boat and car, and a vessel for brain-state measurement in the natural world. Its first ride with a consumer headband shows which of its modes a wet, moving, unpressed headband can measure. The paddle board, yes, with usable EEG on nine epochs in ten. The ball, as motion, with a step rate the accelerometer reads cleanly and EEG bands it inflates. The water, on the forehead only. The theta/beta ratio read 2.7 on the board, in the range of our earlier floating work, and the other two ratios are motion and water rather than mind. The next ride carries a strap, a trigger, the fNIRS stream and the two-minute blocks, and then the question changes from whether the craft can be measured to what it does to the rider.

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